

4.1 $(x_i, y_i, z_i) = (0, 0, 1), (1, 0, 1), (0, 1, 1), (1, 1, 2), (1, -1, 0), (-1, 1, 1)$ v $\mathbb{R}^3$ $z = ax + by + c$, aby $\sum_i (z_i - (ax_i + by_i + c))^2$ było minimalni

$$A = \begin{pmatrix} 0 & 0 & 1 & | & 1 \\ 1 & 0 & 1 & | & 1 \\ 0 & 1 & 1 & | & 1 \\ 1 & 1 & 1 & | & 2 \\ 1 & -1 & 1 & | & 0 \\ -1 & 1 & 1 & | & 1 \end{pmatrix}$$

z twierdzenia 8.91: $A^* A \hat{x} = A^* \vec{b}$,
nad $\mathbb{R}$: $\hat{x} = (A^* A)^{-1} A^* \vec{b}$

$$A^T A = \begin{pmatrix} 0 & 1 & 0 & 1 & 1 & -1 \\ 0 & 0 & 1 & 1 & -1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & -1 & 1 \\ -1 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 4 & -1 & 2 \\ -1 & 4 & 2 \\ 2 & 2 & 6 \end{pmatrix}$$

$$A^T \vec{b} = \begin{pmatrix} 0 & 1 & 0 & 1 & 1 & -1 \\ 0 & 0 & 1 & 1 & -1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \\ 2 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix}$$

$$(A^T A) \hat{x} = A^T \vec{b}: \begin{pmatrix} 4 & -1 & 2 & | & 2 \\ -1 & 4 & 2 & | & 4 \\ 2 & 2 & 6 & | & 6 \end{pmatrix} \sim \begin{pmatrix} 4 & -1 & 2 & | & 2 \\ 0 & 15 & 10 & | & 18 \\ 0 & -5 & -10 & | & -10 \end{pmatrix} \sim \begin{pmatrix} 4 & -1 & 2 & | & 2 \\ 0 & 15 & 10 & | & 18 \\ 0 & 0 & -20 & | & -12 \end{pmatrix}$$

$$\begin{aligned} a &= \frac{\frac{4}{5} - \frac{6}{5} + 2}{4} = \frac{10-2}{5} \cdot \frac{1}{4} = \frac{2}{5} \\ b &= \frac{18}{15} - \frac{10}{15} \cdot \frac{3}{5} = \frac{18-6}{15} = \frac{12}{15} = \frac{4}{5} \\ c &= \frac{12}{20} = \frac{6}{10} = \frac{3}{5} \end{aligned}$$

$\Downarrow$

$$z = \frac{2}{5}x + \frac{4}{5}y + \frac{3}{5}$$